

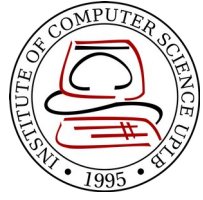
CMSC 137

Data Communications and Networking

Lecture Module
14 – HDLC and PPP

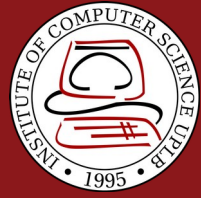


University of the Philippines
LOS BAÑOS



Module 14: HDLC and PPP

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Lecturer



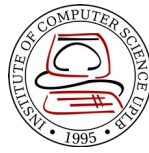
High-level Data Link Control

High-level Data Link Control (HDLC)

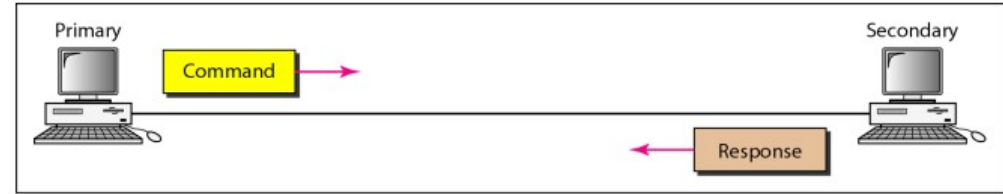


- A bit-oriented protocol for communication over point-to-point and multiple links
- Implements the ARQ mechanisms discussed

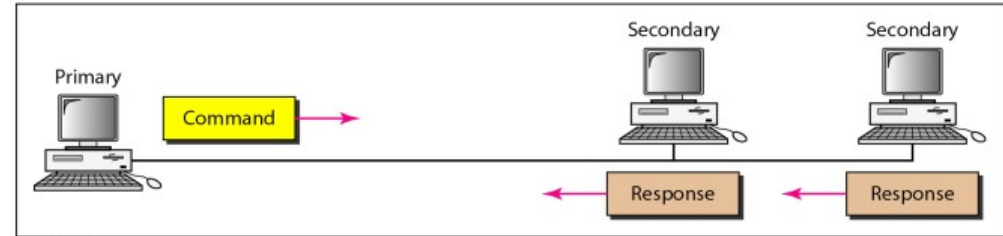
Transfer Modes



- **Normal Response Mode (NRM)** – One primary stations and multiple secondary stations

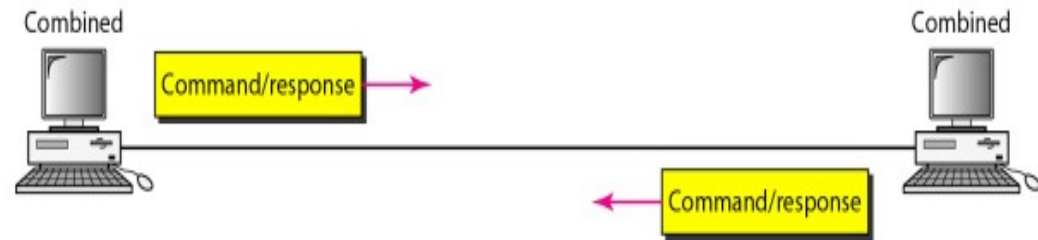


a. Point-to-point



b. Multipoint

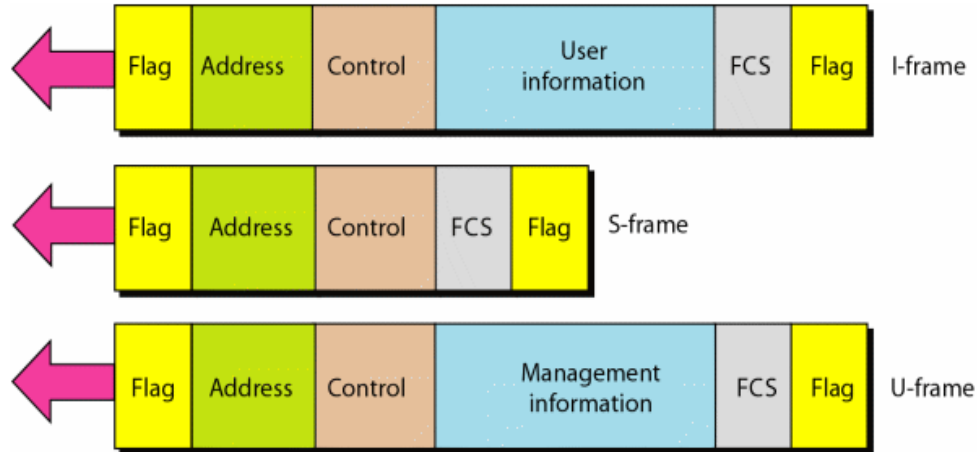
- **Asynchronous Balanced Mode (ABM)** – each station can function as primary and secondary



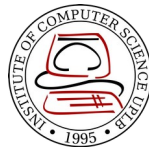
Frames



- Information frames (I-frames) – used to data-link user data
- Supervisory frames (S-frames) - used to transport control information
- Unnumbered frames (U-frames) – reserved for system management

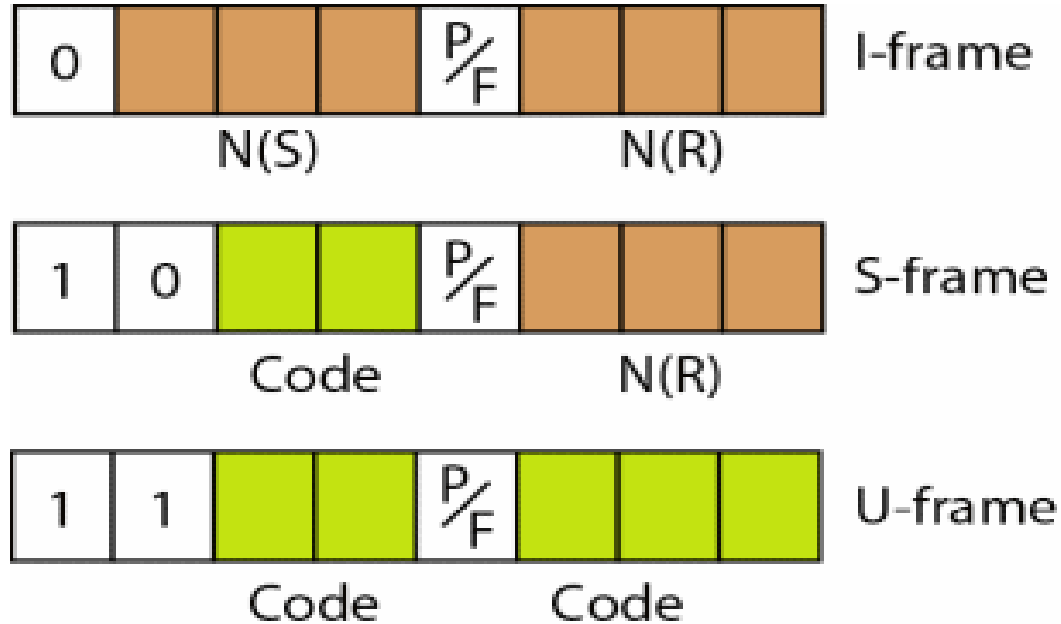
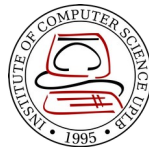


Frame Fields

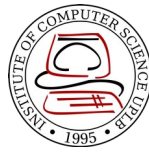


- **Flag** – 8-bit pattern: 01111110
- **Address** – to secondary from primary; from secondary to primary;
 - can be 1 byte or several bytes
 - 1 byte can identify up to 128 stations
 - The last bit of the last byte is always ends with 1; signals the last byte of the address field
- **Control field** – 1- or 2-byte segment used for flow and error control; depends on the type of frame
- **Information field** – data
- **Frame Check Sequence** – 2- or 4-byte ITU-T CRC

Control Field

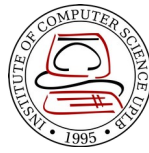


Control Field: I-Frames



- First bit: 0
- $N(S)$ – sequence number of the frame
- $N(R)$ – acknowledgment number when piggybacking is used
- P/F bit – Poll or Final bit

Control Field: S-Frames



- First two bits: 10
- Control Codes
 - Receive ready (RR) - 00
 - Receive not ready (RNR) – 10 ; receiver is busy
 - Reject (REJ) – 01; NAK used in GBN
 - Selective reject (SREJ) – 11; NAK used in SR

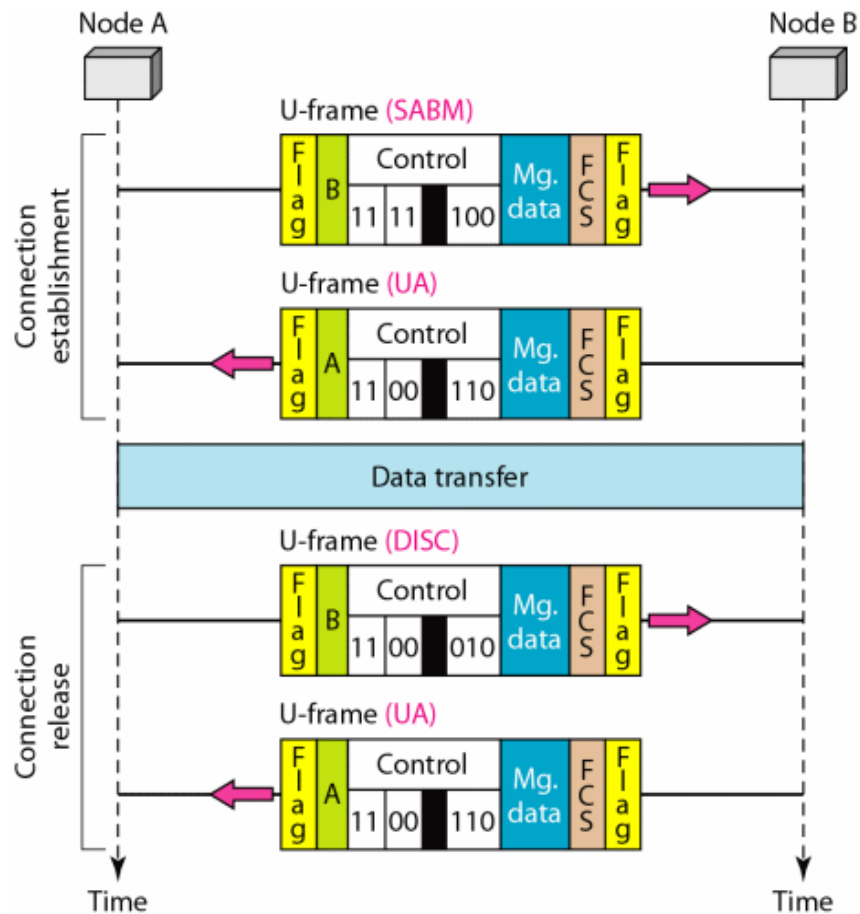
Control Field: U-Frames



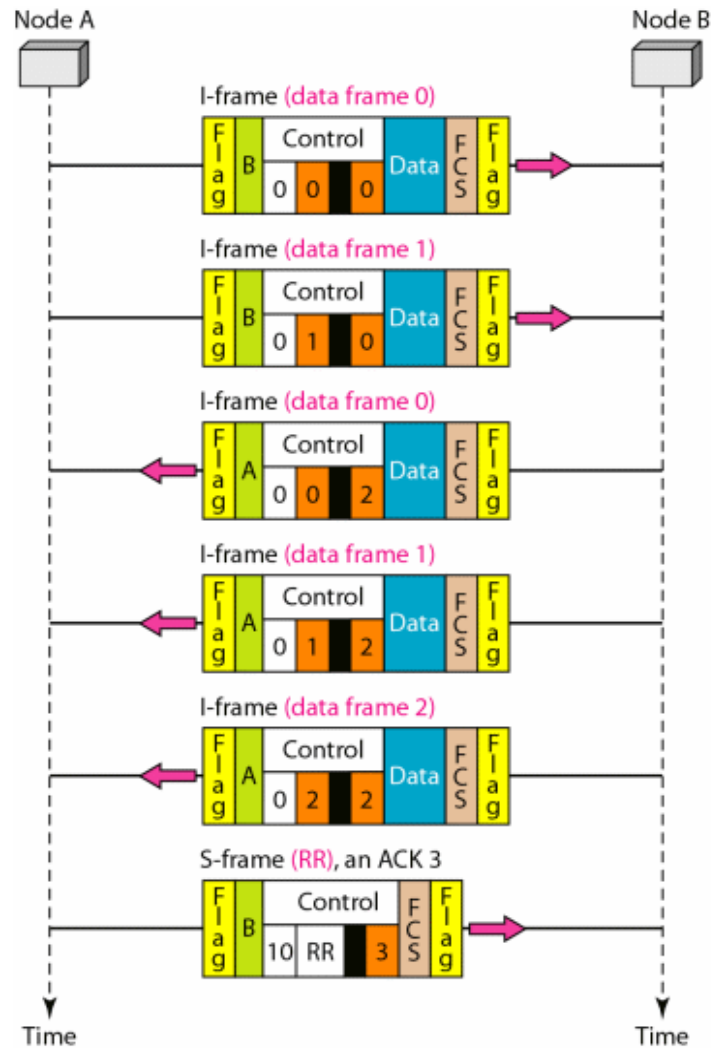
- First two bits: 11

<i>Code</i>	<i>Command</i>	<i>Response</i>	<i>Meaning</i>
00 001	SNRM		Set normal response mode
11 011	SNRME		Set normal response mode, extended
11 100	SABM	DM	Set asynchronous balanced mode or disconnect mode
11 110	SABME		Set asynchronous balanced mode, extended
00 000	UI	UI	Unnumbered information
00 110		UA	Unnumbered acknowledgment
00 010	DISC	RD	Disconnect or request disconnect
10 000	SIM	RIM	Set initialization mode or request information mode
00 100	UP		Unnumbered poll
11 001	RSET		Reset
11 101	XID	XID	Exchange ID
10 001	FRMR	FRMR	Frame reject

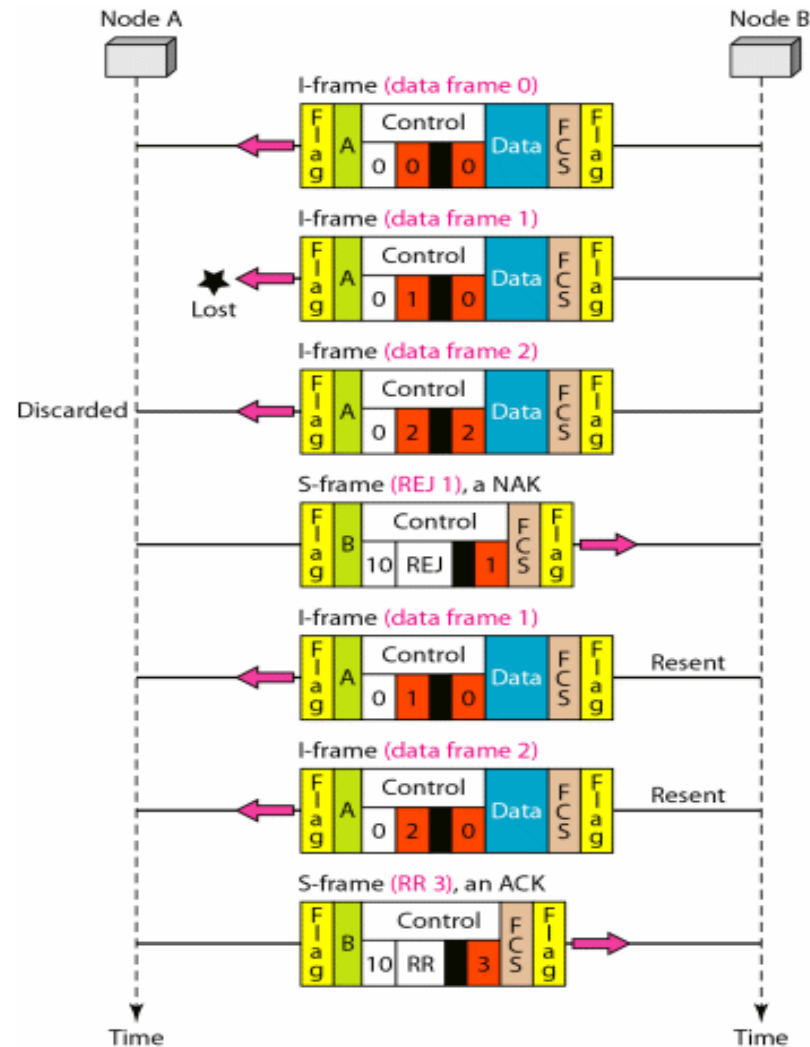
Connection and Disconnection

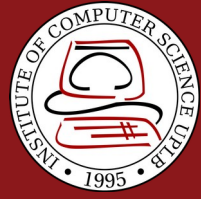


Piggybacking without error



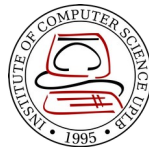
Piggybacking with error





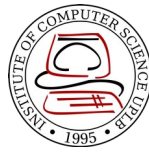
Point-to-Point Protocol

Point-to-Point Protocol (PPP)



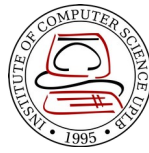
- Used in home computers connecting to ISPs
- Traditional modem connection through a telephone line
- Most common data link layer protocol
- Byte-oriented protocol

PPP Services



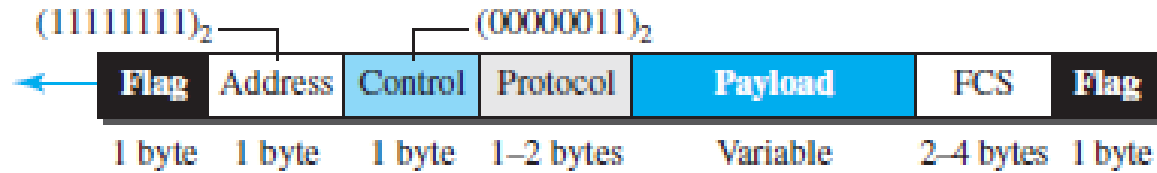
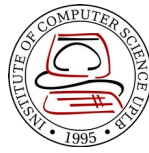
- Defines the format of the frame
- Defines how two devices negotiate link establishment
- Defines how network layer data are encapsulated in the data link frame
- Defines how two devices authenticate each other
- Provides connections over multiple links
- Provides network address configuration

PPP Missing Services



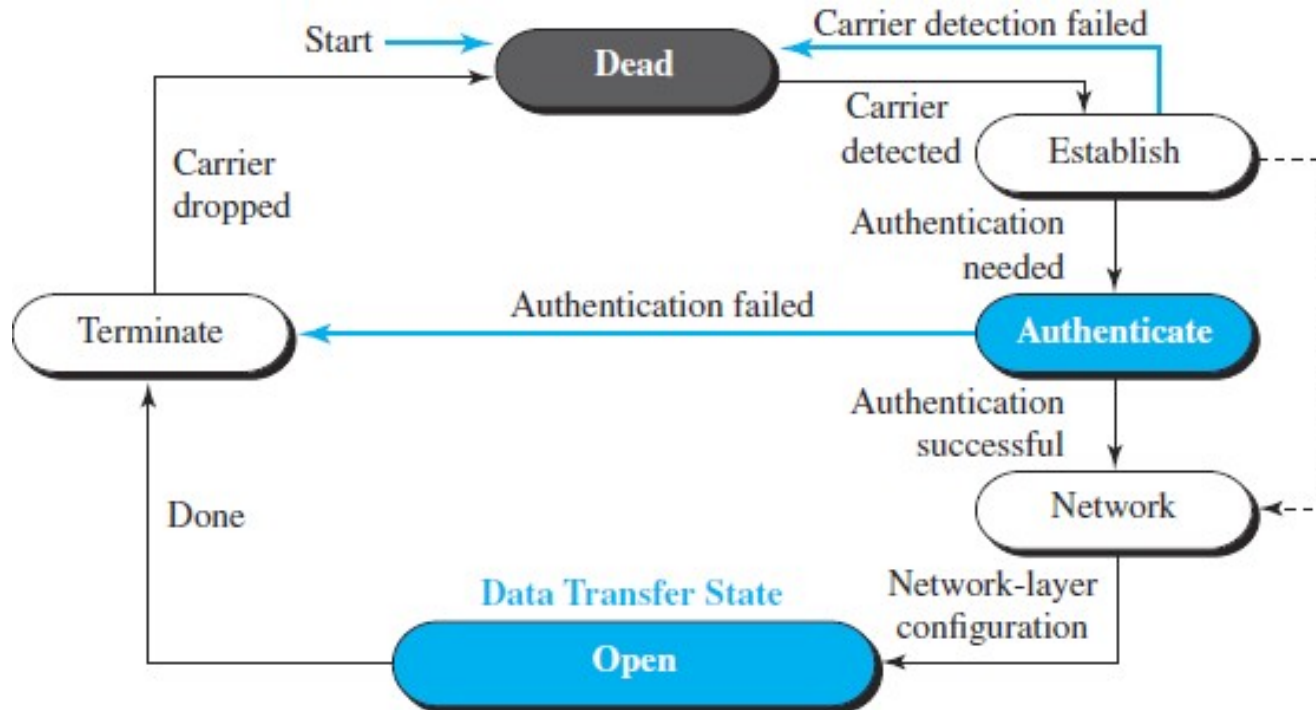
- Does not provide flow control
- Very simple mechanism for error control using CRC
- Does not provide sophisticated addressing mechanism to handle frames in a multipoint configuration

PPP Frame

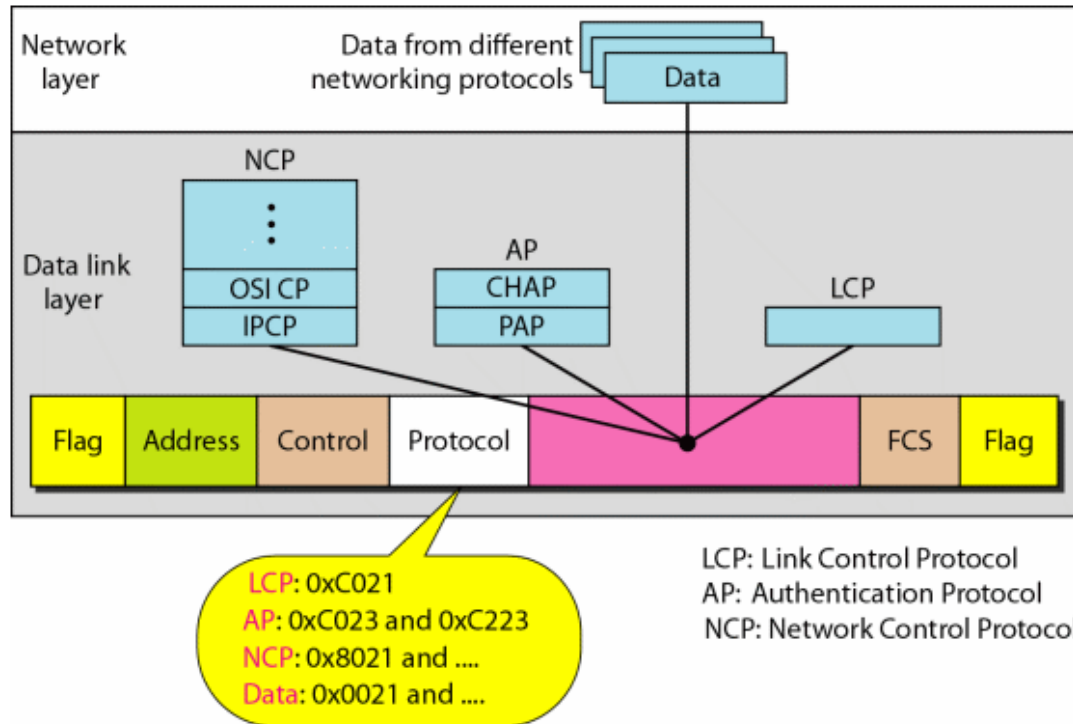


- **Flag** – 01111110; treated as a byte
- **Address** – 11111111 (*broadcast address*)
- **Control** – 00000011 (*imitating unnumbered frames in HDLC*)
- **Protocol** – defines what is being carried in payload
- **Payload** – user data or control information
- **Frame Check Sequence** – 2-byte or 4-byte standard CRC
- **ESC byte** for stuffing: 01111101

PPP Transition Phases



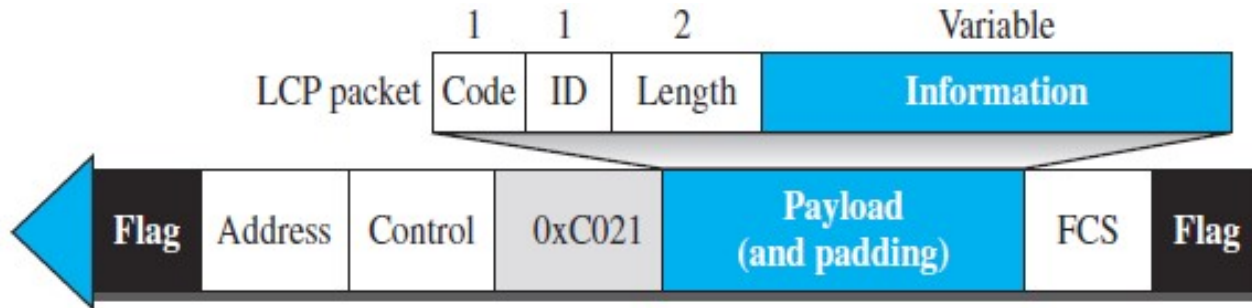
Multiplexing in PPP



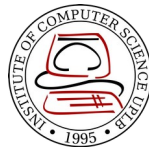
Link Control Protocol (LCP)



- Responsible for establishing, maintaining, configuring and terminating links
- Provides negotiation mechanisms to set options between the two endpoints



Link Control Protocol (LCP)

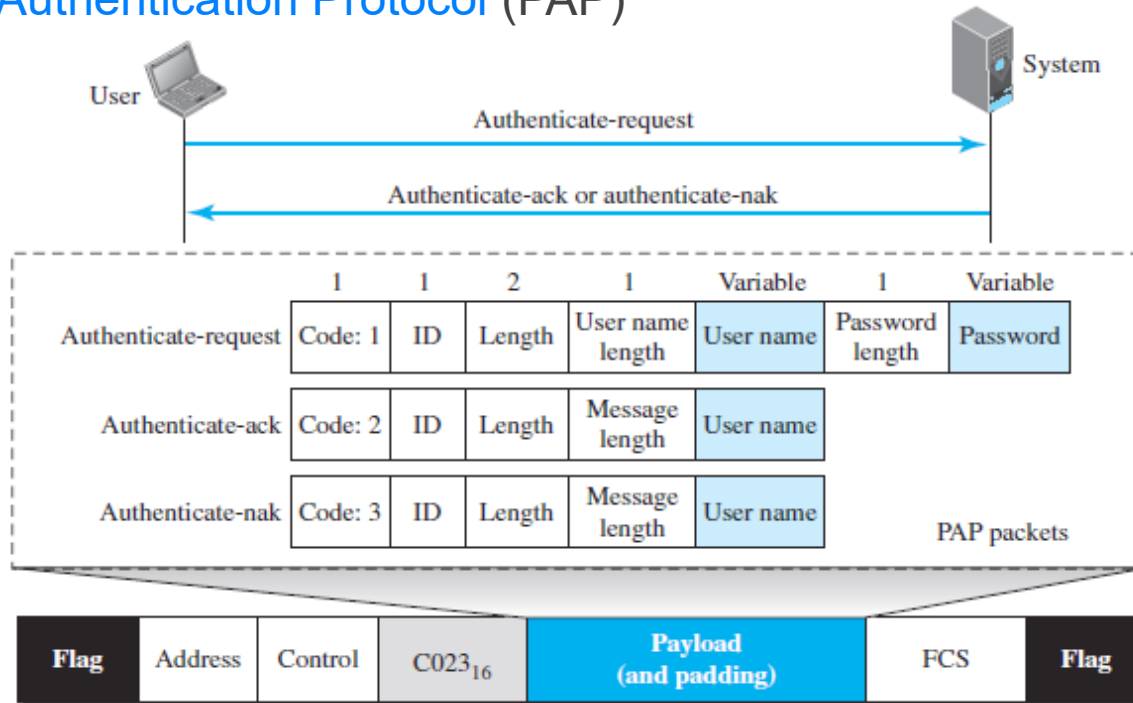


Code	Packet Type	Description
0x01	Configure-request	Contains the list of proposed options and their values
0x02	Configure-ack	Accepts all options proposed
0x03	Configure-nak	Announces that some options are not acceptable
0x04	Configure-reject	Announces that some options are not recognized
0x05	Terminate-request	Request to shut down the line
0x06	Terminate-ack	Accept the shutdown request
0x07	Code-reject	Announces an unknown code
0x08	Protocol-reject	Announces an unknown protocol
0x09	Echo-request	A type of hello message to check if the other end is alive
0x0A	Echo-reply	The response to the echo-request message
0x0B	Discard-request	A request to discard the packet

Authentication Protocols



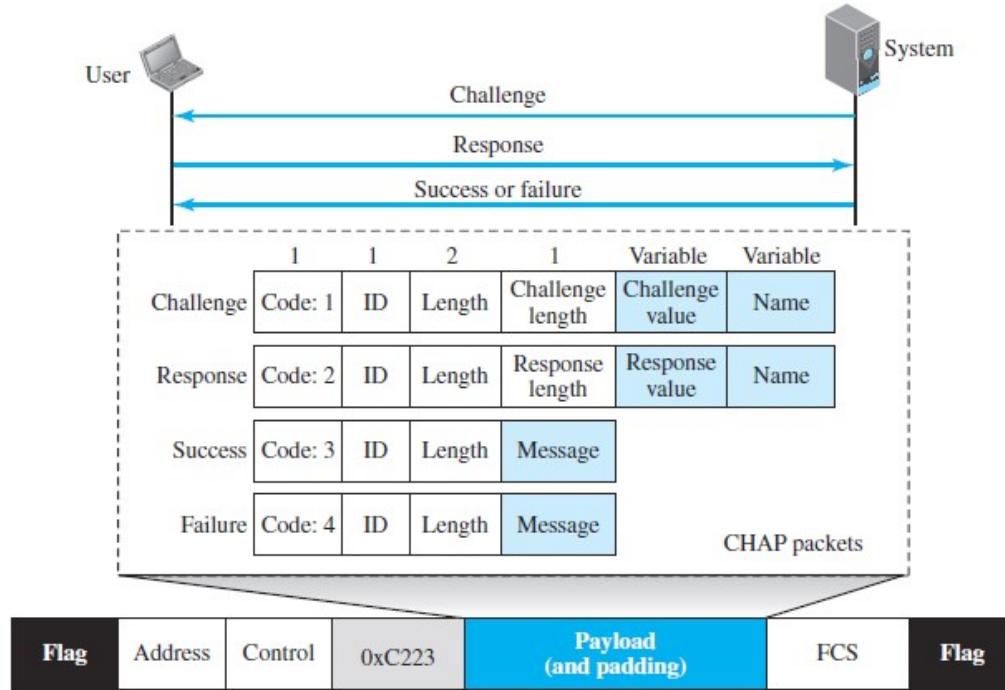
Password Authentication Protocol (PAP)



Authentication Protocols



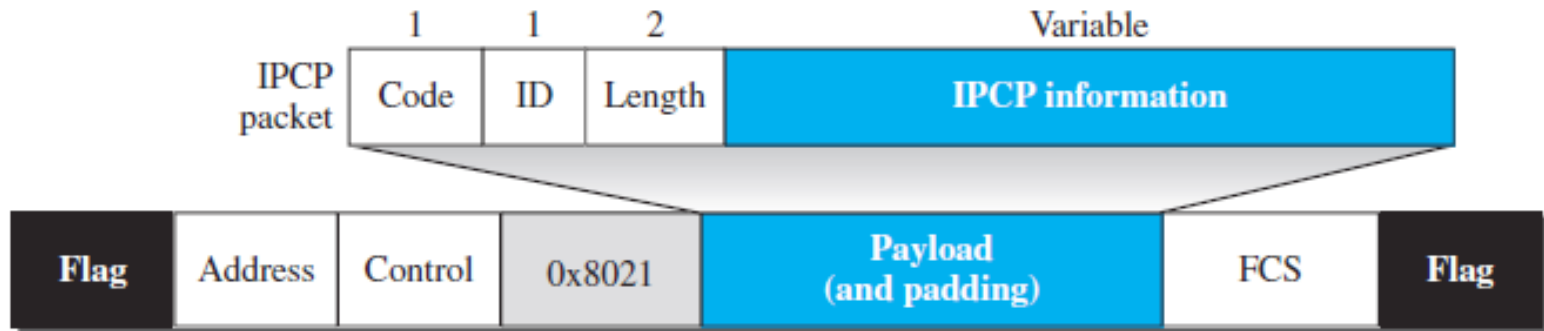
Challenge Handshake Authentication Protocol (CHAP)



Network Control Protocols



- PPP carry a network layer data packet from protocols defined by the Internet
- [Internet Protocol Control Protocol](#) (IPCP) – configures the link used to carry IP packets



References



- Forouzan, B.A. 2013. Data Communications and Networking, 5th Ed. McGraw-Hill, New York.